MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 6278

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, xenon gas

See figure 1

Mounting position: Any

Weight: 5-1/2-ounces (155.9 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Epp	epy	epx	egx	Ib	ib	tk	Ehk	TE	de-ionization
Unit:	V	Vac	v	v	v	Adc	a	sec	Vac	°C	µs
Maximum:	15.4	350	500	500	200	5.0	60	---	100	225	150
Minimum:	12.6 (see note 1)	---	---	---	---	---	---	120	---	---	---
TEST CONDITIONS:	14	350	500	---	---	5.0	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓔ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1201	Short and discontinuity detection	-	No preheat required	0.25	II	---	---	---	---
1231	Peak emission by voltage drop	2	ib = 60 a	0.65	II	etd	---	25	v
1336	Heater-cathode leakage	-	Rg = 10,000 ohms; resistive load; Ehk = 100 Vac; Rhk = 10,000 ohms (max)	0.65	II	Ihk	---	1.0	mA
3201	Critical grid voltage for conduction	-	Ebb = 26 Vdc; Rg = 10,000 ohms	0.65	II	Ecc	---	+20.0	Vdc
(E) 3204	Voltage drop	2,4	Epp = 125 Vac; Ib = 5.0 A	0.65	II	Etd	---	9.5	Vdc
3206	Operation	-	Rg = 10,000 ohms; resistive load	0.65	II	Ecc	+2.0	-7.0	Vdc
3216	Grid current	5	Rg = 1 Meg	0.65	II	Ic	---	5.0	uA
(E) 3214	High-frequency operation	6	Epp = 125 Vac; Rp/Ib = 5.0 Adc; Ecc1 = 22.5 ±2 Vdc; Ecc2 = 45 ±2 Vdc	0.65	II	F	3,400	---	Hz
(E) 3241	Heater current	3	tk = 120 sec (min)	0.65	II	If	2.45	2.70	A
	Quality conformance inspection, part 2								
1031	Variable-frequency vibration	7	t = 300 sec in each plane	---	---	---	---	---	---
---	Heating time (1)	3,8	Ef = 12.6 V; t = 180 sec	---	---	---	---	---	---
---	Heating time (2)	3,9	Ef = 28 V; t = 25 sec	---	---	---	---	---	---
	Quality conformance inspection, part 3								
1516	Stability life	-	Ehk = 100 Vac; TA = room	---	---	---	---	---	---
---	Stability life-test end points:								
3204	Voltage drop	-		---	---	Etd	---	9.5	Vdc
---	Voltage drop change	-		---	---	ΔEtd	---	1.0	Vdc
1506	Heater-cycling life	10	Ef = 28 V; 45 seconds "on" 300 seconds "off"; Ehk = 0; 500 cycles	---	---	---	---	---	---
---	Heater-cycling life-test end points:								
3241	Heater current	-		---	---	If	2.45	2.70	A
1336	Heater-cathode leakage	-		---	---	Ihk	---	1.0	mAdc
1501	Intermittent life	11	Group C; ib = 60 a; Ehk = 100 Vac; TE = 225°C (min)	---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3 -Continued								
---	Intermittent life-test end points (500 hours):								
3206	Operation	-		---	---	Ecc	+2.0	-7.0	Vdc
3216	Grid current	-		---	---	Ic	---	10.0	µA
3204	Voltage drop	-		---	---	Etd	---	10.5	Vdc
3214	High-frequency vibration	-		---	---	F	3,400	---	Hz
1041	Shock	12	300 G	---	---	---	---	---	---
1031	Vibration fatigue	12		---	---	---	---	---	---
---	Shock and vibration-fatigue test end points:								
3206	Operation	-		---	---	Ecc	+2.0	-7.0	Vdc
3241	Heater current	-		---	---	If	2.45	2.70	A
1336	Heater-cathode leakage	-		---	---	Ihk	---	1.0	mA

NOTES:

1. The maximum duration of 28 volts on the heater shall be 45 seconds; the minimum time between applications shall be 5 minutes.
2. The grid shall be connected to the anode through a 10,000-ohm resistor.
3. Heater voltage shall not have been applied in the 5-minute period prior to the commencement of this test.
4. Use an alternating current source and a resistive load. Measure the power loss of the tube with a wattmeter used in such a manner as to have the current coil in series with the tube and the voltage coil across the tube. The Etd is equal to the total power loss minus the coil loss under load, divided by the average of Ib.
5. The tube shall conduct 5 amperes for a minimum of 5 minutes before the grid voltage is increased to cutoff.
6. Test shall be taken between a minimum operating time of 15 seconds and a maximum of 60 seconds.
7. Tube shall not conduct when under vibration with the following conditions applied: Ebb = 125 Vdc minimum; Rg = 10,000 ohms; Ecc = -10 Vdc; Ef = 14.0 V.
8. Conditions of operation shall be applied at end of specified warmup time. The tube shall conduct properly without sparking.
9. At the end of the specified warmup time, switch immediately to 14 V heater and simultaneously apply test condition voltages. The tube shall conduct properly without sparking.
10. When no defectives occur in three lots inspected, sampling may be reduced to approximately every 30 days. A lot failure is cause for requalification for reduced sampling.
11. Heater voltage shall be applied, after the "off" period, 120 seconds before the anode voltage.
12. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with an acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

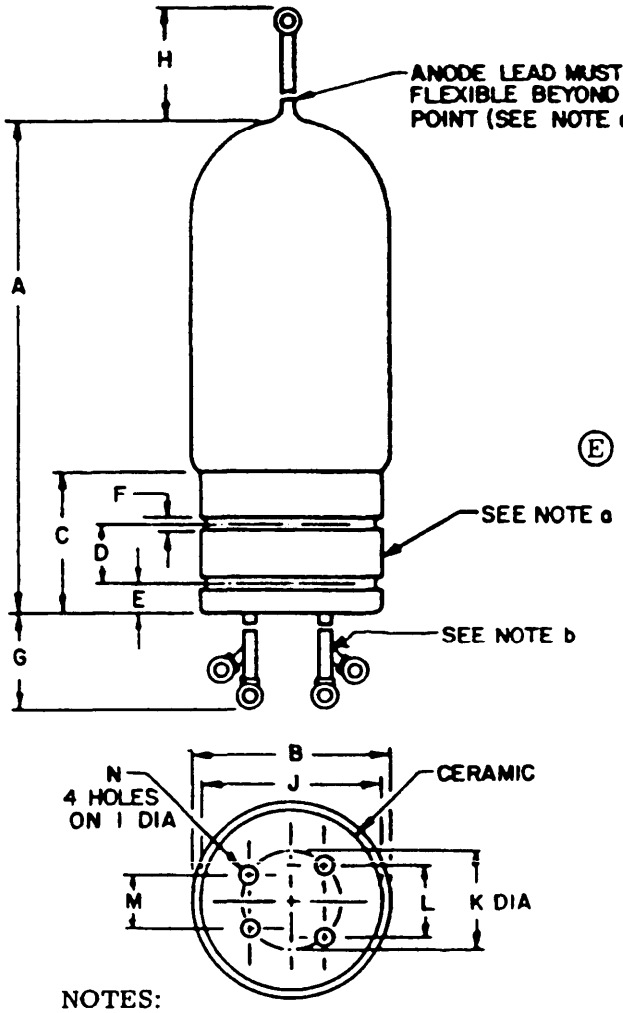
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2938)

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:41Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	---	5.125 (130.18)
B	---	2.063 (52.40)
Quality conformance inspection, part 3 (see note d)		
C	1.410 (35.81)	1.470 (37.34)
D	.620 (15.75)	.630 (16.00)
E	.285 (7.24)	.315 (8.00)
F	.110 (2.79)	.140 (3.56)
G	3.500 (88.90)	---
H	5.500 (139.70)	---
J	1.845 (46.86)	1.875 (47.63)
N	.190 (4.83)	.200 (5.08)
Reference dimensions		
K	1.000 (25.40)	
L	.750 (19.05)	
M	.562 (14.27)	

NOTES:

a. Clamping; caution should be exercised in clamping the tube other than below the upper groove of the base.

b. Flexible leads, silicone impregnated glass fiber insulation, class C-2, or equivalent with lugs (A. M. P. No. 31882, or equivalent) for No. 6 stud. Colors: heater, brown; cathode, yellow; grid, green. Lead sleeving material shall exhibit full color impregnation.

c. Silicone impregnated glass fiber insulation, class C-2, or equivalent with lug (A. M. P. No. 31882, or equivalent). Color: red. Lead sleeving material shall exhibit full color impregnation.

d. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 6278.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:41Z
Check the source to verify that this is the current version before use.